

# YIXUAN ASHER ZHANG

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## EDUCATION

### University of Illinois Urbana-Champaign

Master of Science in Computer Science

Champaign, IL

May 2028 (expected)

**Coursework:** Machine Learning Systems, Distributed Systems, Advanced Operating Systems, Advanced Topics in NLP, Computer Networks, Advanced Computer Security.

### University of Nottingham

B.Sc. (Hons) Computer Science, First Class · GPA 3.9/4.00

Nottingham, UK

Jun 2026

**Honors:** Outstanding Graduate (Top 5%, 2026); Dean's Scholarship (Top 10%, 2025); Outstanding Student (Top 5%, 2025); Zhejiang Provincial Government Scholarship (Top 1%, 2024); MCM Honorable Mention (2024).

**Selected Coursework with 4.0/4.0:** Computer Vision (ranked 1st), Computer Security (ranked 1st), Operating Systems & Concurrency, Algorithms & Data Structures, Artificial Intelligence Methods, Databases and Interfaces, Systems and Architecture.

## SELECTED INTERNSHIPS

### Ningbo AST Machinery Technology | [github.com/asher0913/multi-material-slicer](https://github.com/asher0913/multi-material-slicer)

Ningbo, China

Software Engineer Intern | C++17, Qt, OpenGL, CMake, Python, FastAPI, Vue.js, MySQL, LLMs

May 2026 – Aug 2026

- Built a cross-platform multi-material resin 3D-printing slicer in C++17, Qt and OpenGL: STEP/STL assemblies imported through OpenCascade with nested transforms preserved, per-part material assignment and a VBO-based 3D preview.
- Rasterized per-material PNG layer masks in a background worker with cancellation and timeouts, planned tank changes layer by layer and wrote merged exposures plus G-code; packaged the app for macOS and Windows.
- Added a headless --selftest that runs the full import-to-G-code path under Xvfb in Ubuntu GitHub Actions, alongside CMake builds and unit tests for the Python slicing backend.
- Designed a 51-table MySQL CRM (107 foreign keys, 70 constraints) and 10 Vue/FastAPI modules; built an LLM pipeline that produced 345 faculty profiles and 1,127 deduplicated grant records.

### Zhiyang Innovation Technology

Shanghai, China

Machine Learning Engineer Intern | PyTorch, ONNX, TensorRT, NVIDIA Jetson, FastAPI, Docker

May 2025 – Aug 2025

- Cut Jetson Orin Nano inference latency from 220 ms to 95 ms (2.3×) and model size from 87 MB to 14 MB (6.2×) at under 0.5 mAP loss by exporting to ONNX and deploying INT8 TensorRT FP16 behind FastAPI and Docker.
- Raised mAP@0.5 from 0.74 to 0.79 and recall 6.2 points while cutting false positives 12% on a 7:1 imbalanced defect set; a training pipeline for 120K+ images cut label noise 18% and data preparation from 3 days to under 1, adding 5.6 mAP@0.5.

### Beijing Corefire Technology

Beijing, China

Software Engineer Intern | Java, Spring Boot, Spring Cloud, MySQL, Redis, REST APIs

May 2024 – Jul 2024

- Implemented merchant onboarding and POS/QR query REST APIs for UMFintech in Spring Cloud and MySQL, plus reports that summarized onboarding volume, terminal activations and provider profit sharing by day, month or custom range; built token-based merchant login with sessions stored in Redis.

## SIDE PROJECTS

### SkillRouter | Python, LLM agents, embeddings | [github.com/asher0913/skill-router](https://github.com/asher0913/skill-router)

Aug 2026 – Sep 2026

- Three-tier skill registry with an IDF-weighted lexical prefilter and a ranker: on a 60-skill catalog, per-query context fell from 14,108 to 2,052 tokens (−85.5%) at top-1 accuracy 0.984.

### CodeTaskForge | Python, FastAPI, Docker, pytest | [github.com/asher0913/code-task-forge](https://github.com/asher0913/code-task-forge)

Sep 2025 – Oct 2025

- Built a SWE-bench-style judge for coding-agent patches: path policy, throwaway workspaces, shell-free pytest runs with scrubbed environments and process-group timeouts, restored tests, fail-to-pass / pass-to-pass and held-out tests.
- On 98 labeled candidate patches across 10 tasks, exit codes accepted 25 wrong ones (test tampering, hard-coded answers), the SWE-bench protocol 15 and held-out tests none; GitHub Actions reruns all 294 verdicts.

### Edge Quantization Lab | Python, NumPy, INT8/INT4 PTQ | [github.com/asher0913/edge-quantization-lab](https://github.com/asher0913/edge-quantization-lab)

Apr 2025 – May 2025

- Implemented post-training quantization from scratch (per-channel scales, three calibration observers, sensitivity-driven mixed precision) and an integer-only  $\text{int8} \times \text{int8} \rightarrow \text{int32}$  kernel that matches the simulated model on 100% of predictions.
- Reached 6.7× compression at 97.87% accuracy (FP32 98.18%, 5 seeds); an ablation against the exhaustive optimum exposed a greedy-ordering flaw on 1 of 5 seeds, fixed by ranking layers by KL per byte saved.

**InferenceLab** | *Python, asyncio, vLLM, LLM serving* | [github.com/asher0913/inference-lab](https://github.com/asher0913/inference-lab) Sep 2024 – Oct 2024

- Built an LLM serving front end: an async dynamic batcher, single-flight coalescing of identical in-flight prompts and a TTL/LRU semantic cache keyed by tenant, model and output limit, over a swappable vLLM, SGLang or Ollama backend.
- Coalescing cut backend generations from 16 to 3 in an 80-request, 16-concurrency load test; a labeled paraphrase study caught the semantic cache serving wrong answers, and per-family thresholds answered 67.6% of paraphrases with none wrong.

**Recommendation Ranking Lab** | *Python, SciPy, scikit-learn* | [github.com/asher0913/recsys-ranking-lab](https://github.com/asher0913/recsys-ranking-lab) Jun 2024 – Jul 2024

- Built a two-stage recommender on MovieLens-100K: popularity, ItemKNN and implicit-ALS retrieval (54.3% candidate recall) plus a GBDT re-ranker, evaluated by chronological leave-last-out over the full catalog.
- Lifted HR@10 from 0.097 (ALS) to 0.136 and NDCG@10 from 0.052 to 0.068 over 5 seeds; ablations showed recent-item similarity drives the lift and hard negatives hurt the linear ranker.

**SecureRAG** | *Python, FastAPI, RAG, BM25* | [github.com/asher0913/secure-rag](https://github.com/asher0913/secure-rag) Oct 2023 – Nov 2023

- Multi-tenant hybrid BM25 + embedding retrieval with ACL filters before ranking and a live, fail-closed re-check against the directory of record: after permission changes, a stale-index pre-filter leaked on 54% of queries versus 0% with the re-check.

## SELECTED PROJECTS

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**Oxford-IIIT Pet Classification** | *PyTorch, CNNs* | [github.com/asher0913/oxford-pet-classification](https://github.com/asher0913/oxford-pet-classification) Nov 2024 – Dec 2024

- Fine-tuned an ImageNet ResNet-18 with differential learning rates, cosine scheduling, EMA and flip TTA to 89.8% on the official 3,669-image test split (37 breeds), scoring the test set only once.
- Ran a 10-run progressive ablation of a from-scratch 2.78M-parameter SE-ResNet (35.2% → 49.7%): augmentation added 8.8 points and Mixup cost 1.3; analyzed errors with confusion matrices and a hand-written Grad-CAM.

**Contactless Breathing Monitor** | *Swift, iOS, Vision* | [github.com/asher0913/contactless-breathing-monitor](https://github.com/asher0913/contactless-breathing-monitor) Mar 2024 – Apr 2024

- Synchronized front-camera video with TrueDepth depth and used Vision shoulder landmarks to track the chest on-device; an autocorrelation estimator reports breaths per minute within 1 breath/min on 99.6% of synthetic signals at 60% noise.

**IAMABOT (MechMania 32)** | *Python* | [github.com/asher0913/IAMABOT](https://github.com/asher0913/IAMABOT) Sep 2026

- 24-hour AI contest bot commanding up to 32 units on a 32×32 map, with fire and payload control derived from the game engine's Rust source and a replay-driven regression arena for testing strategy changes.

## SELECTED RESEARCH

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**DualPath-CEM** | *Final-year thesis, PyTorch, privacy ML* | [github.com/asher0913/DualPathCEM](https://github.com/asher0913/DualPathCEM) Nov 2025 – Jun 2026

- Designed dual-path split inference against model inversion: a Gaussian-protected 16×16×16 SlotCEM tensor carries privacy, a compact MobileNetV3 token carries utility, and calibrated logit fusion combines them; the attacker sees both tensors.
- On a 44,139-image FaceScrub split (526 identities, 5 seeds), reached 81.96% mean top-1, 1.63 points above the published Noise\_ARL+CEM reference, while raising decoder and GAN reconstruction MSE 75.8% and 74.9%.
- Released an audited snapshot of 70 formal attack runs and 15 utility repeats with a data manifest; a CI check verifies every reported metric against the frozen results.

*Manuscript in preparation: Y. Zhang and J. Ren, "DDP: A Disentangled Dual-Path Privacy-Protection Framework for Collaborative Face Inference," Elsevier Pattern Recognition, 2026.*

## SELECTED EXTRACURRICULAR

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**Students' Union, University of Nottingham** Nottingham, UK  
*Deputy Director* May 2023 – Apr 2024

- Organized a multilingual cultural season and film festival (400+ participants) and partnered with KPMG, NIO, Xiaomi, PwC, ByteDance and Lufthansa on career sessions, each engaging 200+ students.

**School of Computer Science, University of Nottingham** Nottingham, UK  
*Peer Mentor* Sep 2023 – Jan 2024

- Mentored 20+ first-years in weekly Java, recursion, data structures, complexity and Git sessions; authored review sheets and earned helpful ratings from 90%+ of mentees.

## TECHNICAL SKILLS

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**Languages:** Python, C++17, Java, SQL, TypeScript, JavaScript, Go, Swift

**ML / AI:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LLMs, RAG, AI agents, LoRA fine-tuning, embeddings, computer vision, NLP, OpenCV

**ML Systems:** vLLM, LLM serving, ONNX, TensorRT, INT8 quantization, MLflow, Docker, Kubernetes, CI/CD (GitHub Actions)

**Backend & Tools:** FastAPI, REST APIs, Spring Boot, MySQL, PostgreSQL, Redis, AWS, GCP, Linux, Git, Qt, OpenGL, CMake